

Quick guide for WHC

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Introduction

The WHC features enables the dialog between the operator module and a PLC via a BUS interface. Specific frames enable to display various elements like ASCII text or bitmap images wherever and whenever on the operator module screen.

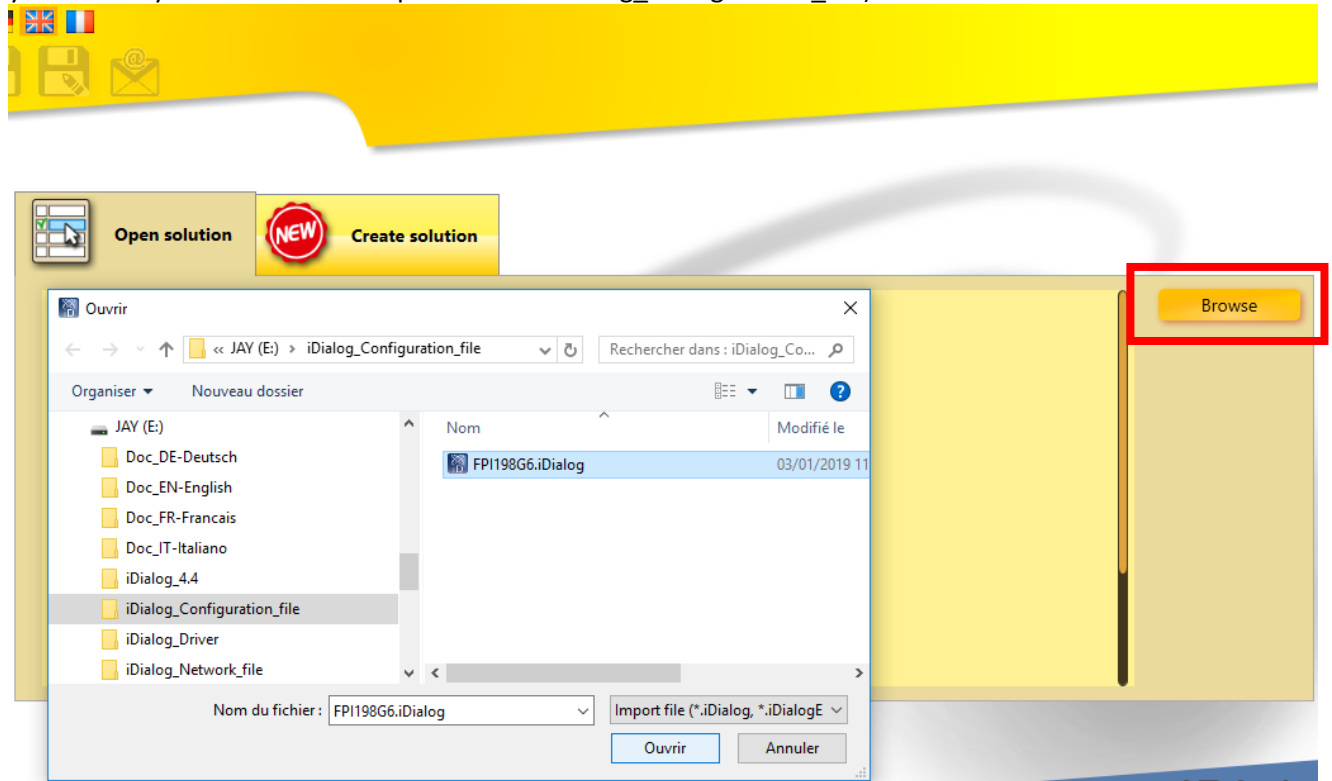
With this function your operator module can truly be a Man/Machine Interface for your PLC.

Any receiver with a BUS communication can use WHC function.

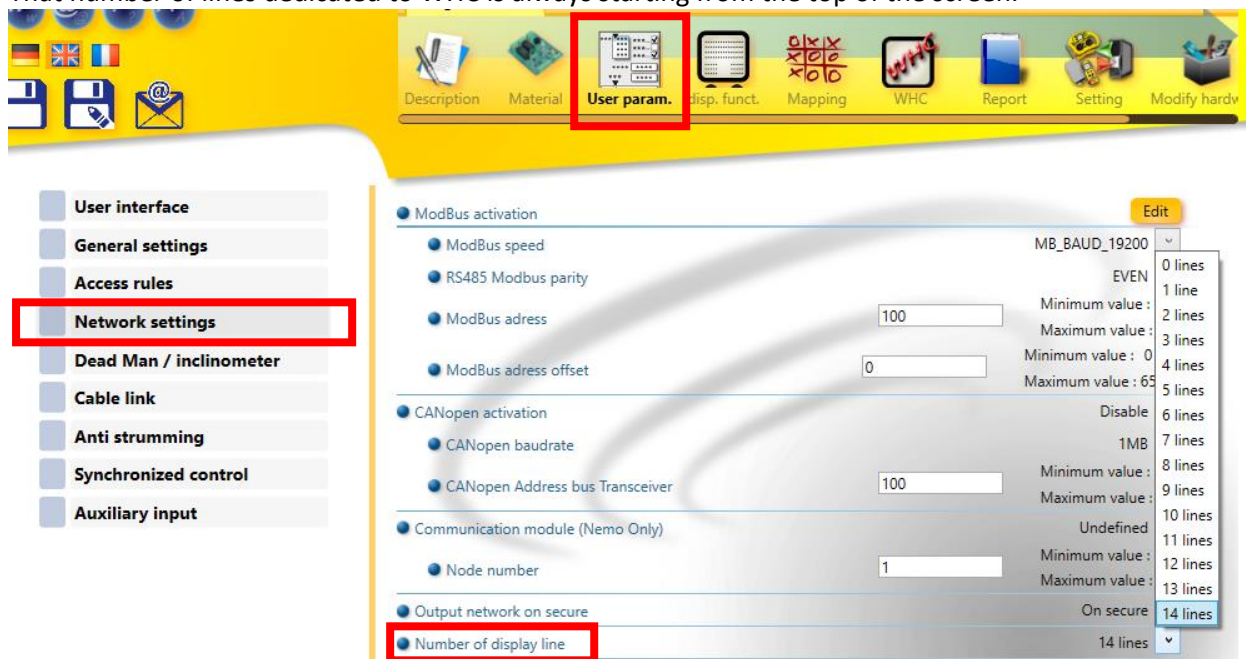
WHC enabling in your project

Network configuration

First of all, open iDialog software, then open your configuration file with it (which you can find on your USB key delivered with the product into iDialog_Configuration_file).



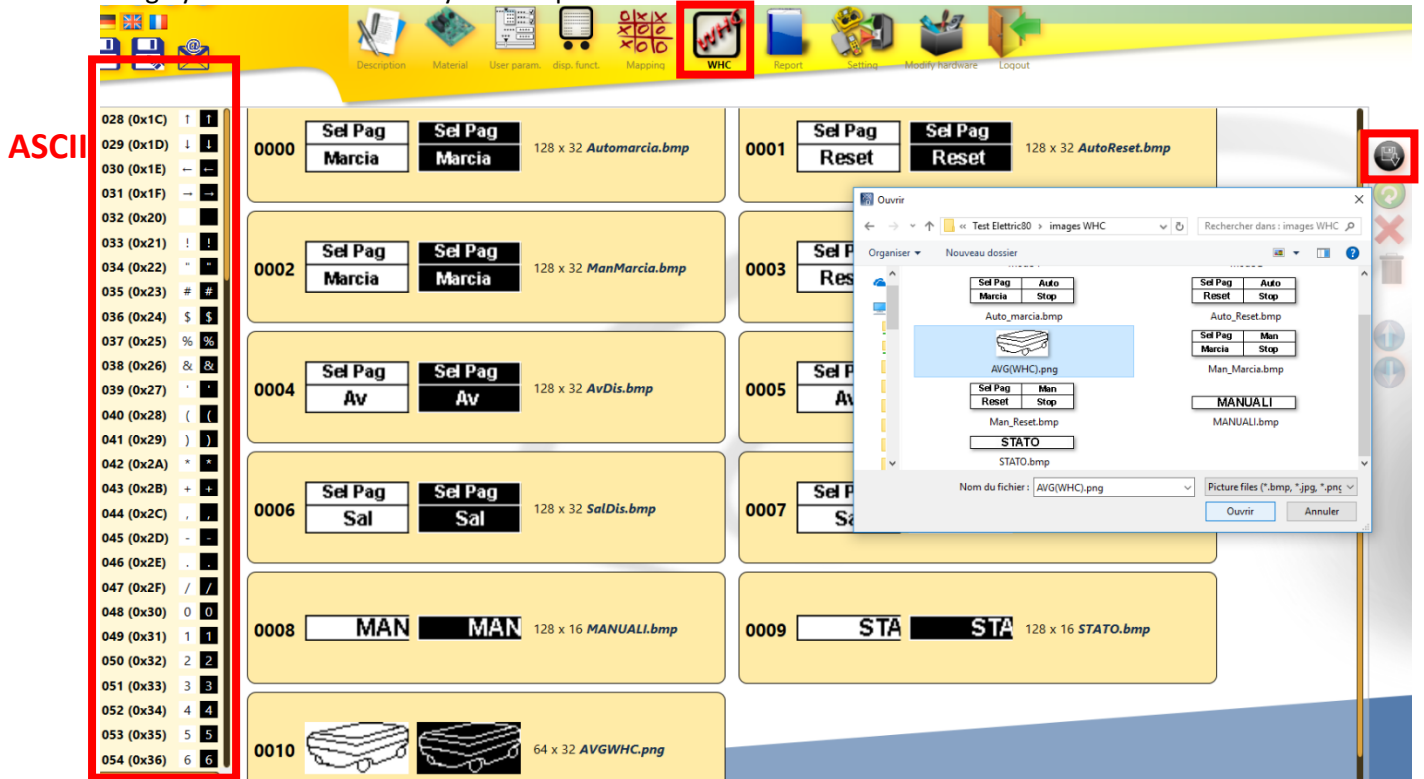
Then go into “User Parameters” > “Network Settings” and activate it by choosing a “number of display line” superior to 0 (full screen dedicated to WHC would be 14 lines, half screen would be 7). That number of lines dedicated to WHC is always starting from the top of the screen.



Don't forget to activate your bus communication (Modbus or other) on the same screen if not already activated.

Upload WHC images into your project

As said earlier, WHC allows you to print on your screen either ASCII character or Bitmap images. ASCII characters are already all available in the system but if you want to upload images into your program, you will need to go into "WHC" then click on the floppy disk icon in black and select the image you want to insert on your computer.



Characteristics of images you can insert are as following:

- 96 images maximum by project (or 87 552 pixels)
- 4096 pixels maximum by image (height and width modulo 8 pixels):
 - o 128x32
 - o 96x40
 - o 64x64
 - o Etc...
- Monochrome
- BMP / JPEG / PNG

Communication with the PLC

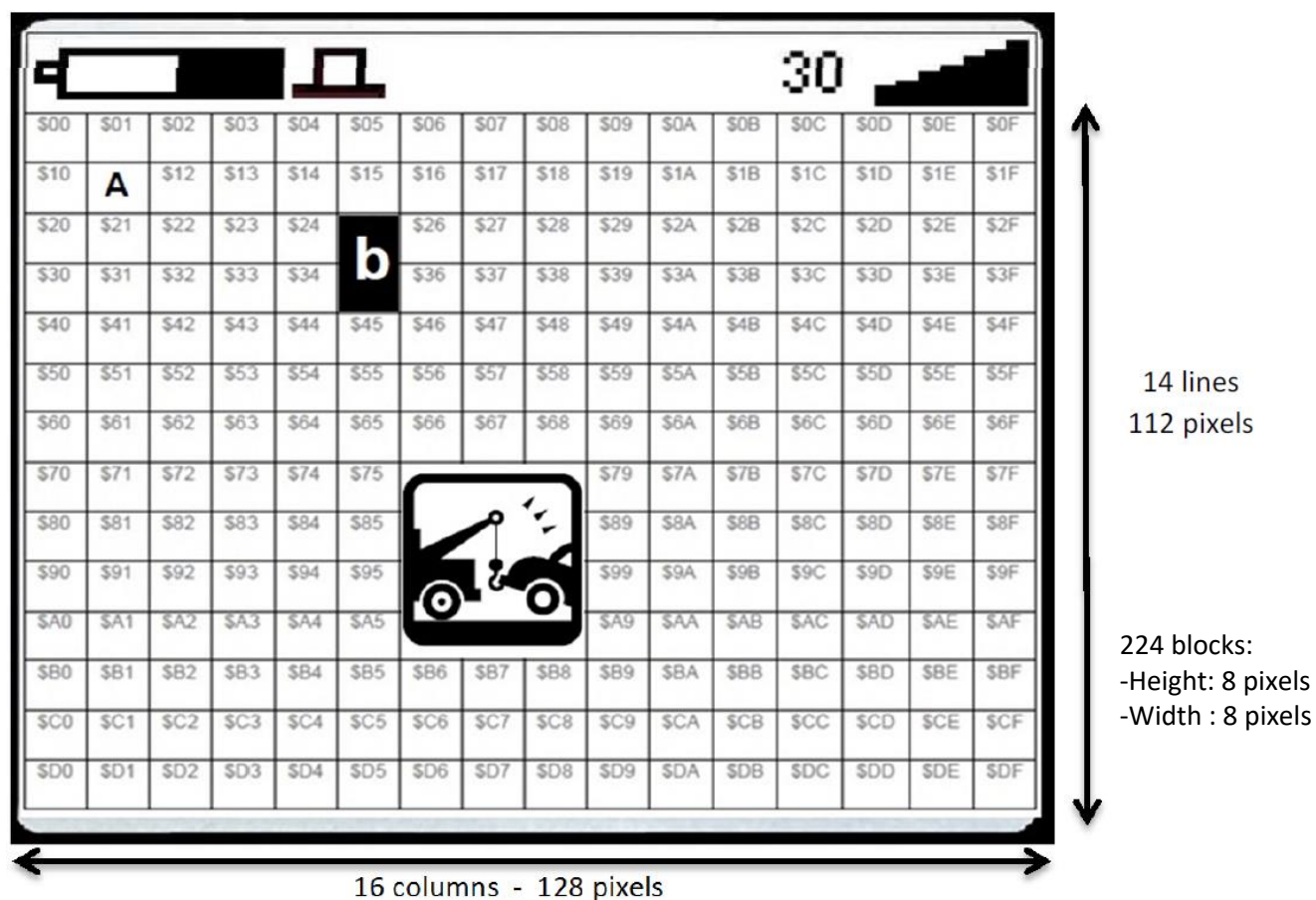
Frame Construction

From now on, the PLC will be the one to handle the WHC display on your transmitter.

The instruction will be given by the PLC through 2 32bytes word with the following frame:

b31...b24	b23...b16	B15...b12	b11...b8				b7...b0
Not Used	Display adress on screen	Not Used	Specific options				Image to display (adress inside your iDialog program)
			Blink	Black&White Inversion	Double Height	ASCII(0) or BMP(1)	
0000.0000	0001.0001	0000	0	0	0	0	0100.0001

With the example frame just above (which is "0x00110041" in hexadecimal), you will then print the ASCII character A onto the adress \$11 as shown below.



The ASCII character "b" and the image that you can see above can be displayed with the following frames:

b31...b24	b23...b16	b15...b12	b11...b8				b7...b0
Not Used	Display adress or block) on screen	Not Used	Specific options				Image to display (adress inside your iDialog program)
			Blink	Black&White Inversion	Double Height	ASCII(0) or BMP(1)	
0000.0000	0010.0101	0000	0	1	1	0	0110.0010 
0000.0000	0111.0110	0000	1	0	0	1	0000.0011 

Registers adress

Address of registers to write in will be these ones:

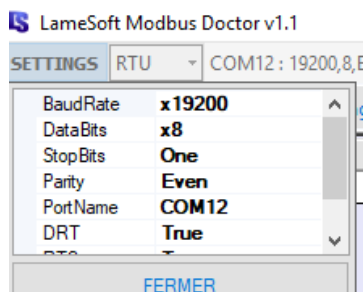
BUS	1 st word (32 bytes) address	2 nd word (32 bytes) address
Modbus RTU	0x0014 to 0x0015	0x0016 to 0x0017
Modbus IP	0x0014 to 0x0015	0x0016 to 0x0017
PROFIBUS	0x0044 to 0x0047	0x0048 to 0x004B
PROFINET	0x0044 to 0x0047	0x0048 to 0x004B
CAN Open	SDO: 0x6412 SubIndex1	SDO: 0x6412 SubIndex2
ETHERCAT	0x20CA SubIndex1	0x20CA SubIndex2
POWERLINK	0x20CA SubIndex1	0x20CA SubIndex2
ETHERNET IP	Param2020	Param2021
DEVICENET	Output21	Output22

Be careful, depending on the bus used, there may be an offset applied on the addressing

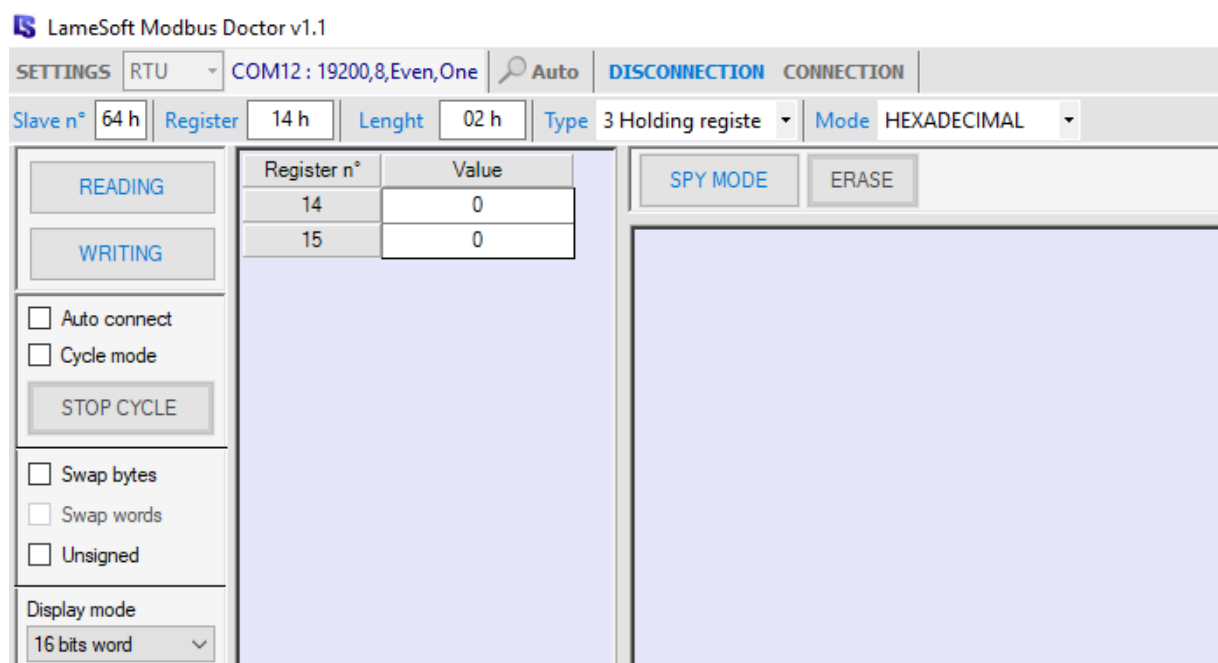
Example

The following example will be made using ModbusDoctor software in order to simulate the PLC sending orders to the JAY receiver.

The JAY receiver used is a TIMO configured with a Modbus RTU communication with 14 lines (full screen) WHC availability.



For all the following steps of this example, I will write into the 16 bytes registers 0x14 and 0x15 which are the 2 registers corresponding to the first WHC 32bytes word.



Register 15 is for Most Significant Bytes : 0b31 to 0b16

Register 14 is for Least Significant Bytes : 0b15 to 0b00

To follow next steps, remember that the frame is as following

Register 15		Register 14			
b31...b24	b23...b16	B15...b12	b11...b8		b7...b0
Not Used	Display adress on screen	Not Used	Specific options		Image to display (adress inside your iDialog program)
			Blink	Black&White Inversion	
			Double Height	ASCII(0) or BMP(1)	



Starting Configuration

As everything on screen will be handled by WHC, we will keep a very simple screen.



For this example, iDialog File is configured with following images in the WHC menu.

Programming

Descriptif Matériel Param. util. Fonc. affich. Mapping WHC Rapport Paramétrage Modification Sortir

028 (0x1C) 1	0000	Sel Pag Marcia	Sel Pag Marcia	128 x 32 Automarcia.bmp	0001	Sel Pag Reset	Sel Pag Reset	128 x 32 AutoReset.bmp
029 (0x1D) 1	0002	Sel Pag Marcia	Sel Pag Marcia	128 x 32 ManMarcia.bmp	0003	Sel Pag Reset	Sel Pag Reset	128 x 32 ManReset.bmp
030 (0x1E) 1	0004	Sel Pag Av	Sel Pag Av	128 x 32 AvDis.bmp	0005	Sel Pag Av	Sel Pag Av	128 x 32 AvInd.bmp
031 (0x1F) 1	0006	Sel Pag Sal	Sel Pag Sal	128 x 32 SalDis.bmp	0007	Sel Pag Sal	Sel Pag Sal	128 x 32 SalInd.bmp
032 (0x20) 1	0008	MAN	MAN	128 x 16 MANUALI.bmp	0009	STA	STA	128 x 16 STATO.bmp
033 (0x21) 1	0010			64 x 32 AVGWHC.png				

First Step : Print the screen at start-up
Print the mode you're in ("Stato")

SETTINGS RTU COM12 : 19200,8,Even,One Auto

Slave n° 64 h Register 14 h Lenght 02 h Type

READING

WRITING

Register n°	Value
14	109
15	C0



You can see that as soon as the first image is printed, everything else of the normal screen is disappearing. We will come back to this matter later.

Then, print the actions corresponding to your buttons just above

SETTINGS RTU COM12 : 19200,8,Even,One Auto

Slave n° 64 h Register 14 h Lenght 02 h Type

READING

WRITING

Register n°	Value
14	100
15	80



As there are 2 possible registers to write WHC, for the same result, you could also have sent two orders at the same time using register 0x14 and 0x15 and also 0x16 and 0x17. For example, you could have printed the mode name and the list of action at the same time.

SETTINGS RTU COM12 : 19200,8,Even,One Auto

Slave n° 64 h Register 14 h Lenght 04 h Type

READING

WRITING

☐ Auto connect

Register n°	Value
14	100
15	80
16	109
17	C0

Second Step: Change the name of the action when pressing a button

Press F1 button and change “Marcia” to “Reset”

SETTINGS RTU COM12 : 19200,8,Even,One Auto

Slave n° 64 h Register 14 h Lenght 02 h Type

READING

WRITING

Register n°	Value
14	101
15	80



Press N2 button and change “Auto” to “Man”

SETTINGS RTU COM12 : 19200,8,Even,One Auto

Slave n° 64 h Register 14 h Lenght 02 h Type 3

READING

WRITING

Register n°	Value
14	103
15	80



Press F1 button and change back “Reset” to “Marcia”

SETTINGS RTU COM12 : 19200,8,Even,One Auto

Slave n° 64 h Register 14 h Lenght 02 h Type

READING

WRITING

Register n°	Value
14	102
15	80



Third Step: Change the mode (list of actions)

Press "Sel Pagina" N1 button and change the mode name AND the list of actions possible corresponding to your new mode.

SETTINGS RTU COM12 : 19200,8,Even,One Auto

Slave n° 64 h Register 14 h Lenght 02 h Type

READING WRITING

Register n°	Value
14	108
15	C0

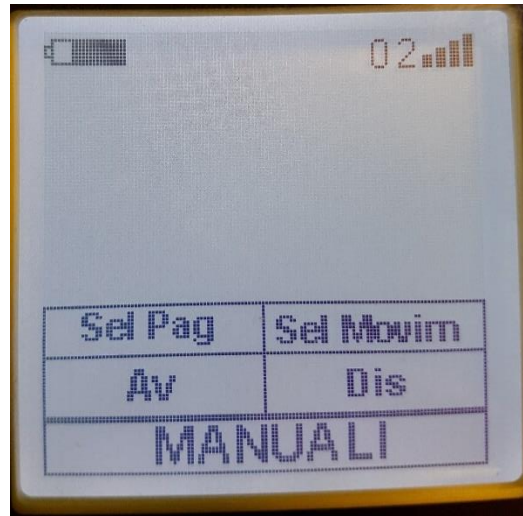
+

SETTINGS RTU COM12 : 19200,8,Even,One Auto

Slave n° 64 h Register 14 h Lenght 02 h Type

READING WRITING

Register n°	Value
14	104
15	80



Fourth Step: Put a title to your page

WHC allows us to make some more graphic design for the screen of our application (on condition to have some design skills !).

On this application, we are driving several small AGV, we will then put a title on screen with of the number of the AGV driven.

SETTINGS RTU COM12 : 19200,8,Even,One Auto

Slave n° 64 h Register 14 h Lenght 02 h Type

READING WRITING

Register n°	Value
14	10A
15	2



To print the number of the AGV, we will use ASCII character 2 and 1 and use the double height characteristic. (double height is not available for images)

SETTINGS RTU COM12 : 19200,8,Even,One Auto

Slave n° 64 h Register 14 h Lenght 04 h Type

READING WRITING

☐ Auto connect

Register n°	Value
14	232
15	1B
16	231
17	1D



Erasing elements

Erasing one part of the screen :

Select the beginning address of the image/letter you want to remove and put an empty image instead (below image 7F (n°127)).

SETTINGS RTU COM12 : 19200,8,Even,One Auto

Slave n° 64 h Register 14 h Lenght 04 h Type

READING

WRITING

☐ Auto connect

Register n°	Value
14	17F
15	1B
16	17F
17	1D



If the address is not the beginning address of the image, this command will do nothing.

Erasing the whole screen :

The ASCII character \$7F is used for erasing the whole screen in one command.

Adress needs to be \$00.

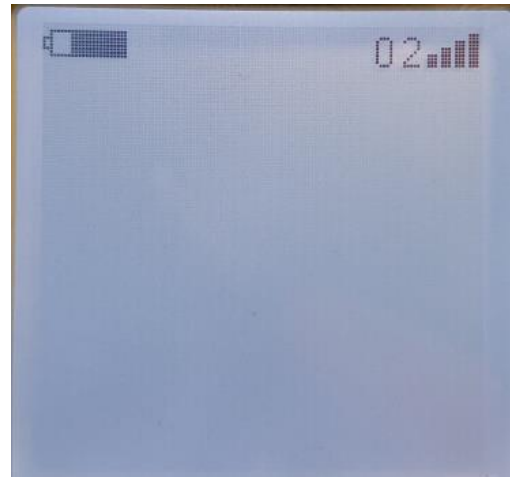
SETTINGS RTU COM12 : 19200,8,Even,One Auto D

Slave n° 64 h Register 14 h Lenght 02 h Type 31

READING

WRITING

Register n°	Value
14	7F
15	0



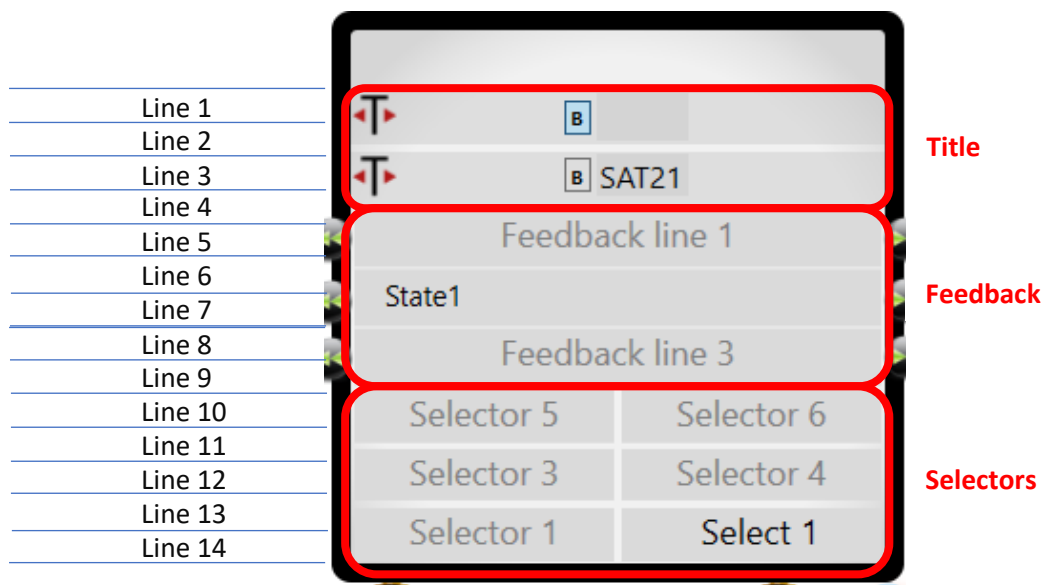
WHC and normal screen superposition

Principle

When using WHC, no matter where you print your image, it will erase the whole normal screen and put your image instead. After that, the only parts that are refreshing of the screen are the parts which values are changing.

That means for parts of the normal screen:

- Title is a fix value sent only once at the beginning:
 - o After the first image (or letter) printed on screen using WHC, the title will disappear
 - o **It will never appear again** (unless the receiver is restarted), you will then need to print a new title using WHC if needed.
- Feedbacks (analogical/logical/...) on the middle of the normal screen:
 - o After the first image (or letter) printed on screen using WHC, it will first disappear, and then reappear above the WHC image.
 - o **This kind of feedback will always be above the WHC.**
- Selectors at the bottom of the normal screen:
 - o After the first image (or letter) printed on screen using WHC, it will disappear.
 - o It will reappear above the WHC image when the value of this selector is changing (for example a button F1 is pressed and change its value)



This is for the case where you are using the full screen WHC configuration (14 lines).

If you don't want your selectors to disappear, you can activate WHC only for the 9 first lines for example.



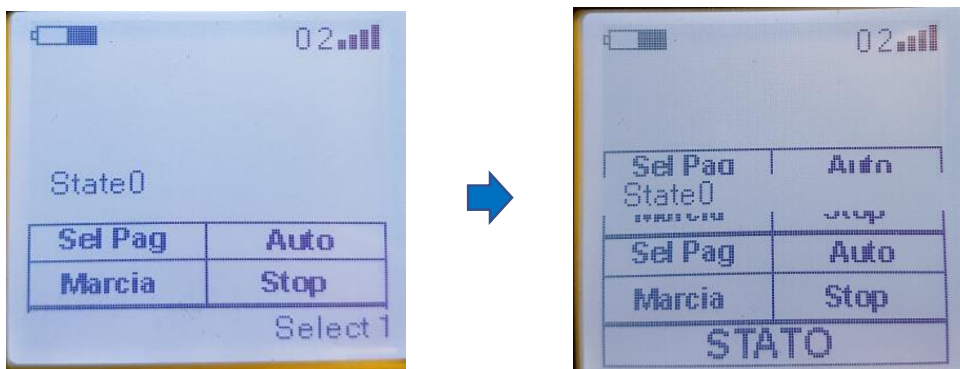
Example



As you can see, title has disappeared whereas the image printed is not at all on it. The selector has also disappeared. Only the feedback is remaining.



By pressing F2, the value of the selector is changed. We can see that its label is appearing again above the WHC.



By making an image appear above the feedback, this one is quickly disappearing but is almost instantly reappearing above the image. You can also see that the selector label has disappeared again. It will only reappear at its next state changing.